

Topology Optimization of electrical machines @ Siemens

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About me

- Academic background
 - Grenoble Applied Mathematics and Computer Science Engineering School (2020)
 - CIFRE PhD – Free Shape / Topology optimization of Electrical Machines (2024)
- Industrial background
 - Hired as scientific developer @ Altair
 - Siemens acquisition
- Today
 - Improve Topology Optimization capabilities in Flux
 - Create a complete tool for electrical machines designers

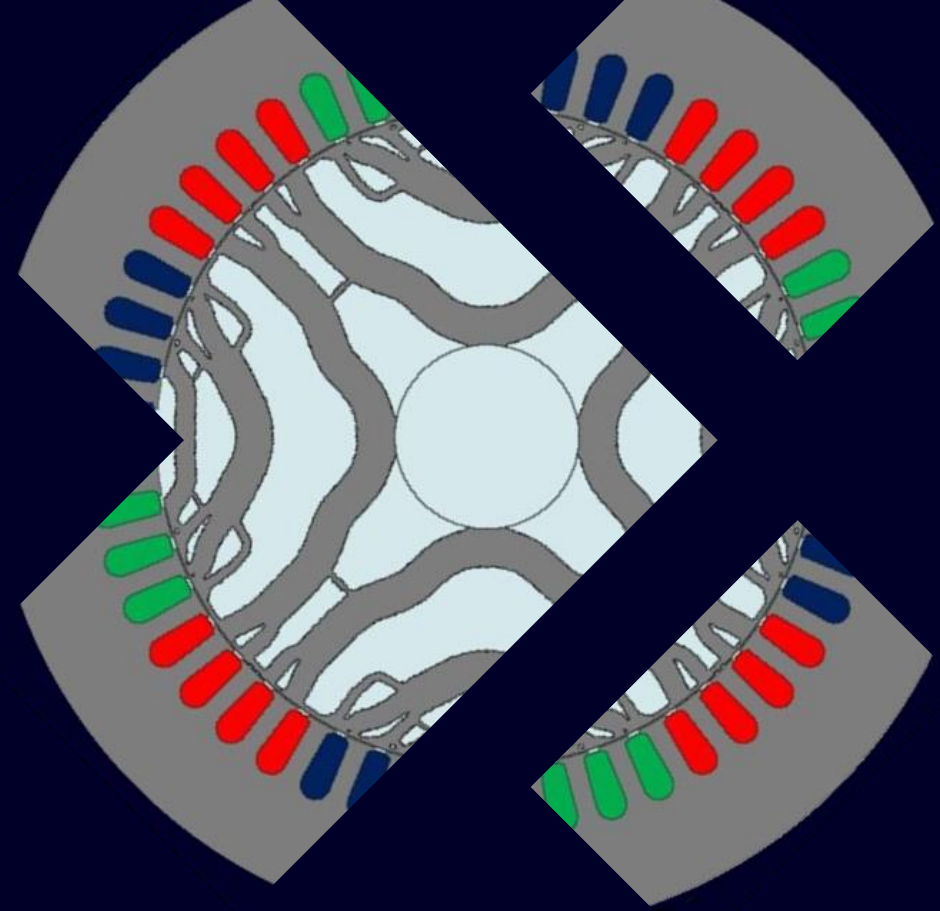


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Topology Optimization of Electrical machines @ Siemens

1. Introduction
2. Numerical solutions
3. Results
4. Simcenter SimLab

Introduction



Structural Optimization methods for electrical machines

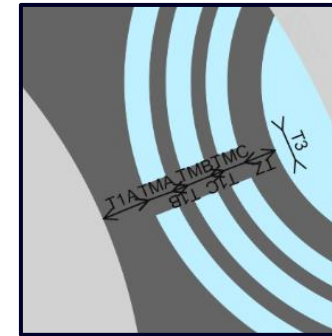
Motor
specifications

OPTIMIZATION

Optimal
geometry

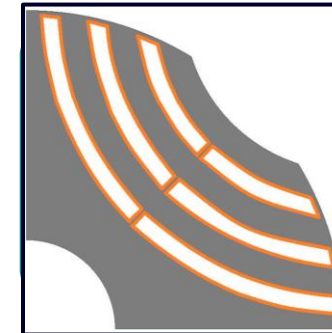
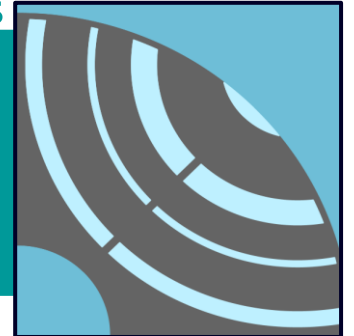
Tradeoff between

- Numerical complexity
- Design exploration & possible innovation
- Manufacturability
- Industrial maturity



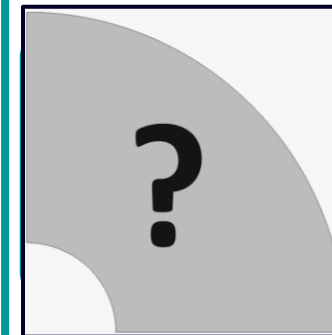
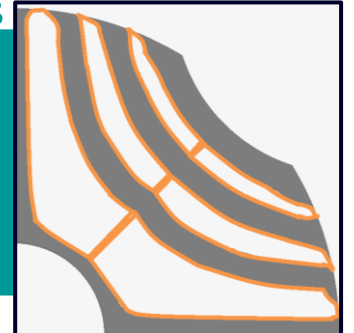
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Parameter
optimization



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Shape
optimization

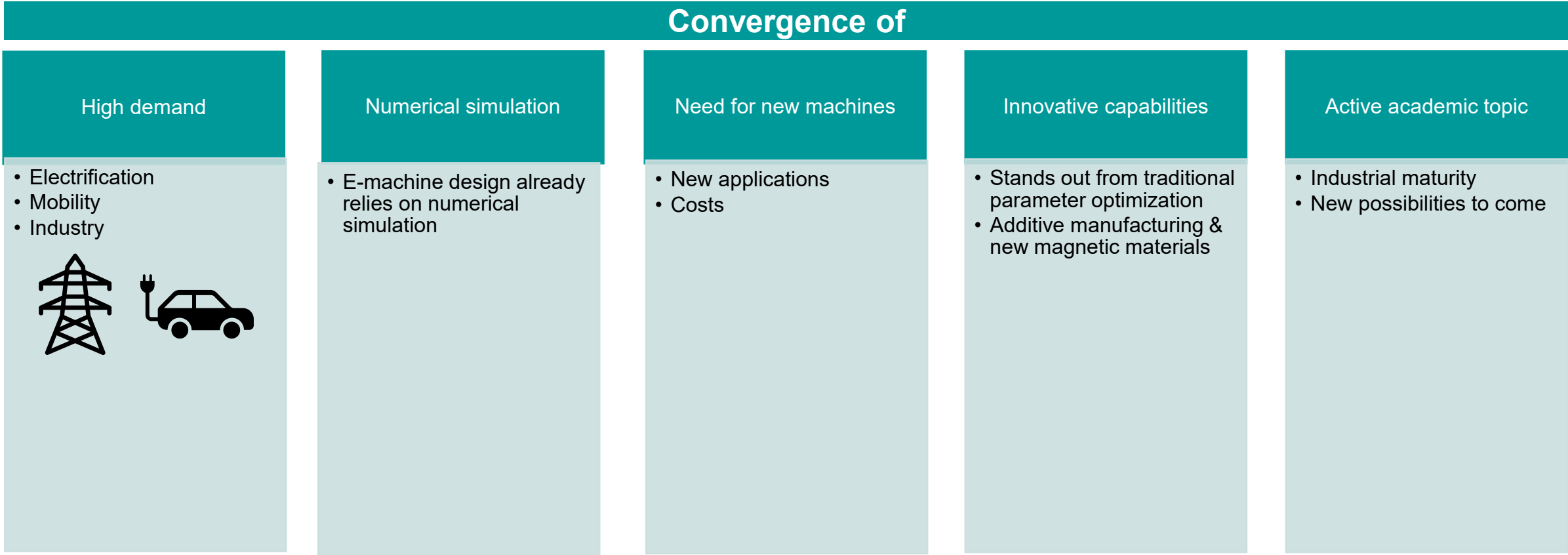


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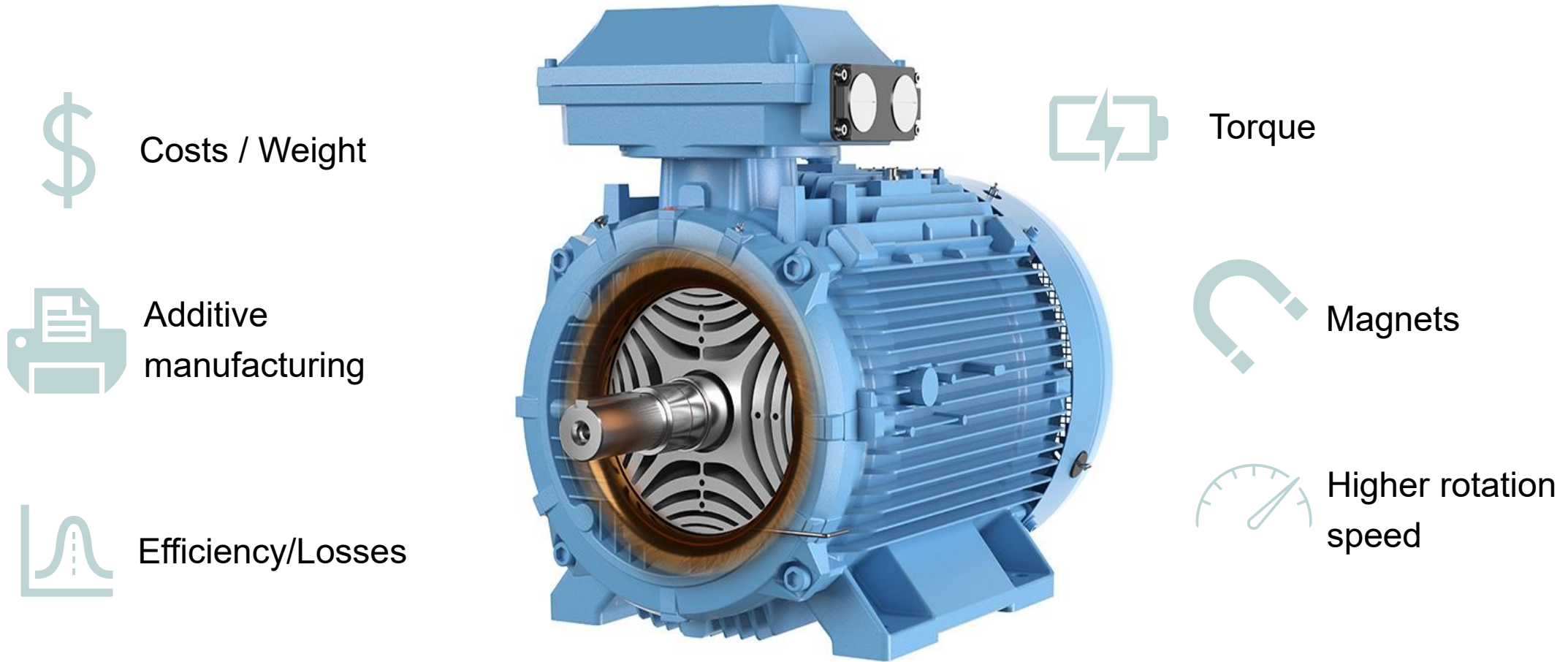
Topology
optimization



Why does Siemens believe in Topology Optimization for Electrical Machines ?

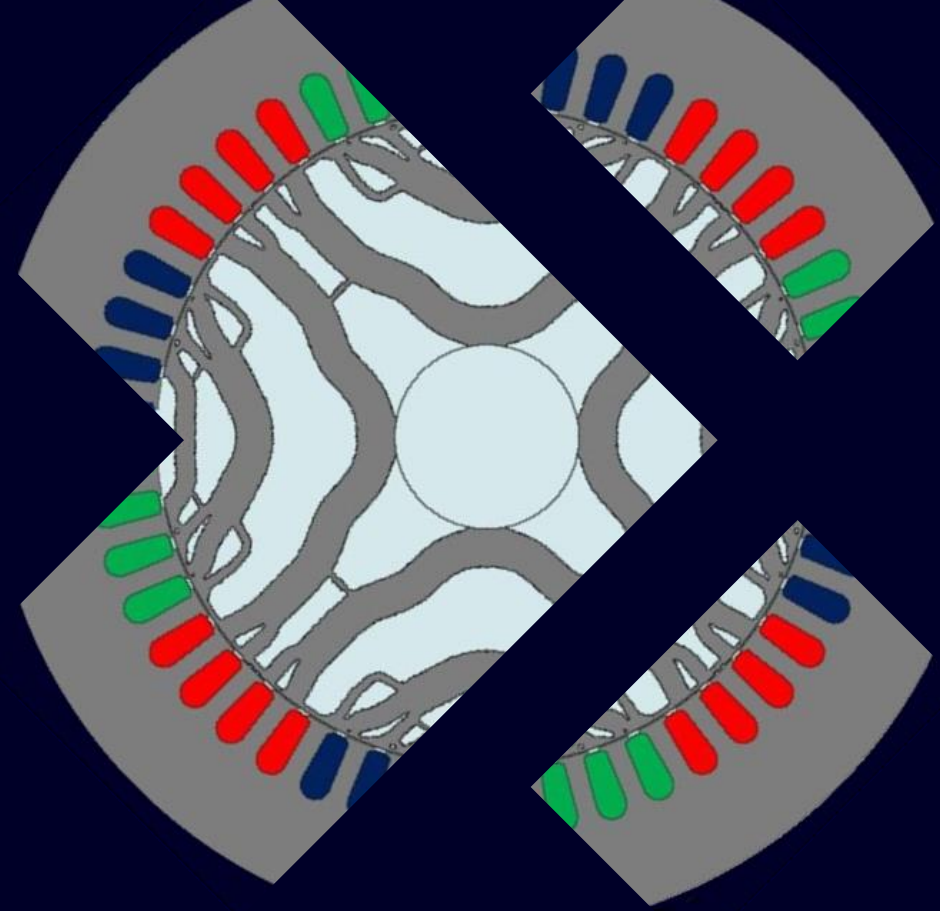


What do the electrical motor designers expect to do with topology optimization?



Topology Optimization of Electrical Machines

Numerical aspects



Siemens (former Altair) Tools for Electromagnetic Topology Optimization



SimCenter Flux

Low-frequency **electromagnetic** solver



Grenoble, France
~40 years old

- Motor finite element analysis
- EM performance computation

SimCenter
OptiStruct



Mechanical solver
Structural Optimizer

Irvine, CA, USA
~40 years old

- Gradient-based topology optimization methods for mechanical applications
- Industrial options & settings

EM Topology Optimization

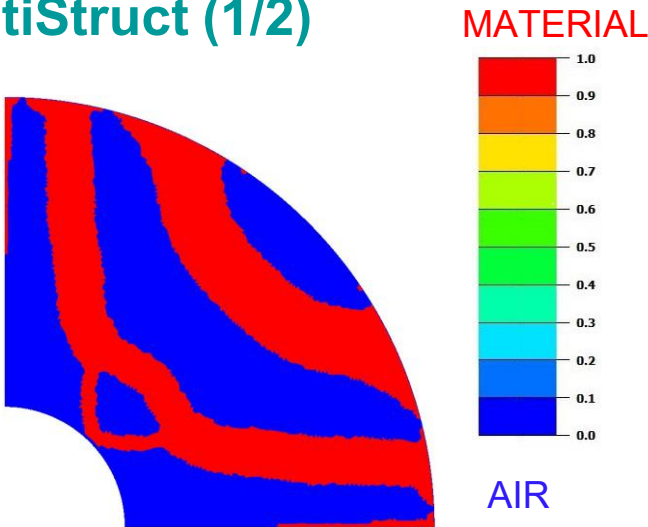


- TO methods in OptiStruct ?
- How to link external TO methods to Flux ?
- Computation of the EM criterion gradients ?

Topology Optimization methods in OptiStruct (1/2)

Density-based approach

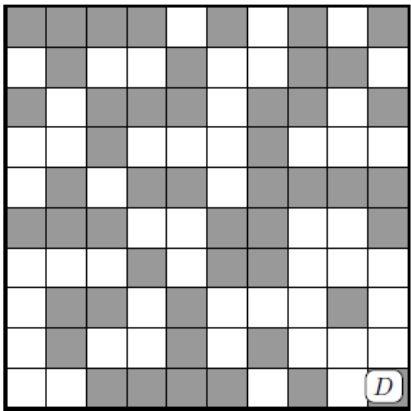
$$\left\{ \begin{array}{ll} \rho = 1 & \text{MATERIAL} \\ \rho = 0 & \text{AIR} \end{array} \right.$$



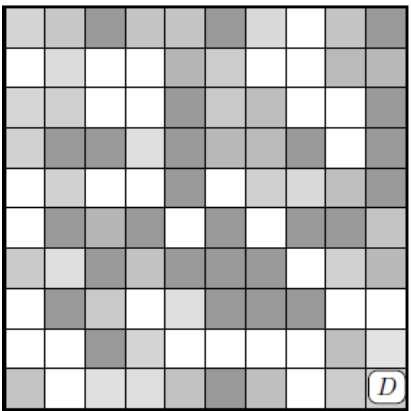
Interpolation of material law ξ

- $\hat{\xi} = \xi_0 + \rho(\xi - \xi_0)$

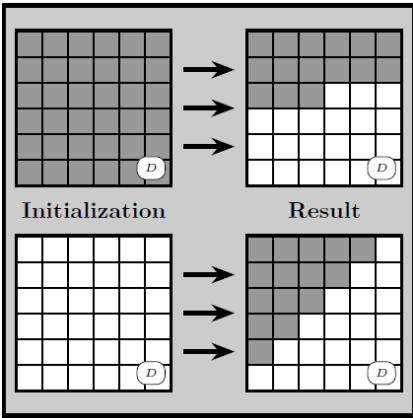
Ill-posed problem



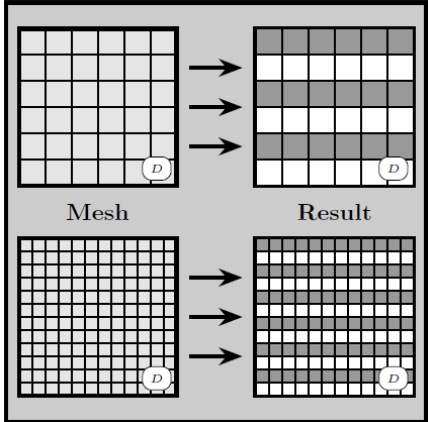
Checkerboard



Intermediate density values



Local minima



Mesh dependency

Topology Optimization methods in OptiStruct (2/2)

Method 1: Guest's approach

Guest, J. K., Prévost, J. H., & Belytschko, T. (2004). Achieving minimum length scale in topology optimization using nodal design variables and projection functions. *International journal for numerical methods in engineering*, 61(2), 238-254.

Method 2: Pseudo-levelset approach

Andreasen, C. S., Elingaard, M. O., & Aage, N. (2020). Level set topology and shape optimization by density methods using cut elements with length scale control. *Structural and Multidisciplinary Optimization*, 62(2), 685-707.

Wang, S., & Wang, M. Y. (2006). Radial basis functions and level set method for structural topology optimization. *International journal for numerical methods in engineering*, 65(12), 2060-2090.

Gradient descent methods

CONLIN (Fleury, C. CONLIN: An efficient dual optimizer based on convex approximation concepts. *Structural Optimization* 1, 81–89 (1989).

<https://doi.org/10.1007/BF01637664>)

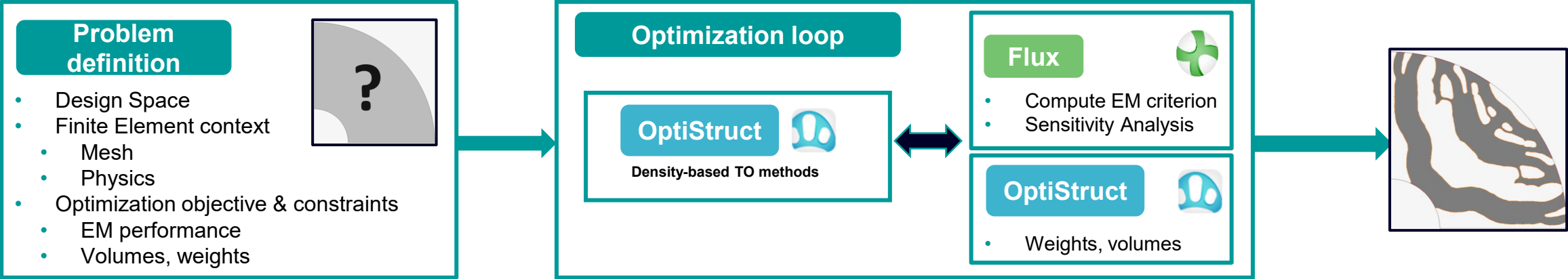
Method of Moving Asymptotes

Method of feasible directions, ...

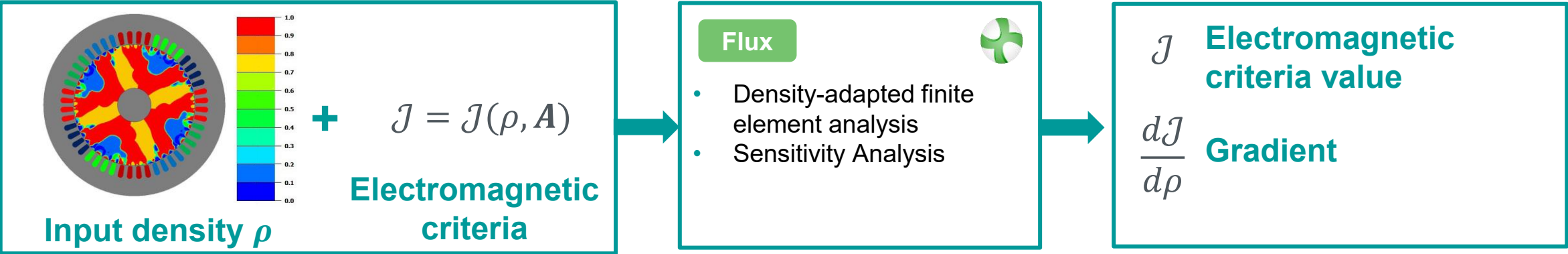
Manufacturing constraints

How to link a density-based TO method to Flux ?

General algorithm



Flux perspective



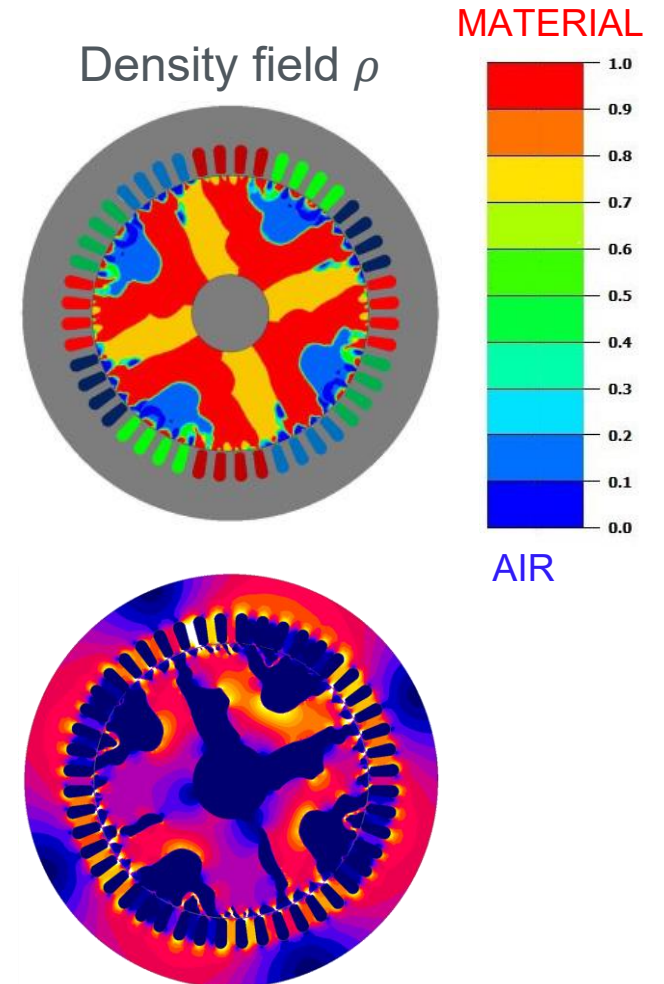
Density-based electromagnetic topology optimization

- 2D magnetostatic analysis
- Density field coming from OptiStruct
- Adapted finite element solving
- Design space made of a ferromagnetic material (magnetic reluctivity $\nu = \nu(\|\mathbf{B}\|)$)
- Interpolation law $\hat{\nu} = \nu_0 + g(\rho)(\nu(\mathbf{A}) - \nu_0)$
- ν_0 : reluctivity in vacuum
- g : penalization operator

$$\begin{cases} \mathbf{B} = \nabla \times \mathbf{A} \\ \nabla \times \mathbf{H} = \mathbf{J} \\ \mathbf{H} = \hat{\nu} \mathbf{B} \\ \mathbf{A} = 0 \end{cases} \quad \begin{matrix} \text{in } \Omega \\ \\ \text{on } \partial\Omega \end{matrix}$$



Implemented in
SimCenter Flux



General sensitivity analysis of Electromagnetic criterion

Adjoint method in a (non-linear) magnetostatic context

Criterion $\mathcal{J} = \mathcal{J}(\rho, A) = \int_{\Omega} j(\rho, A) d\Omega$

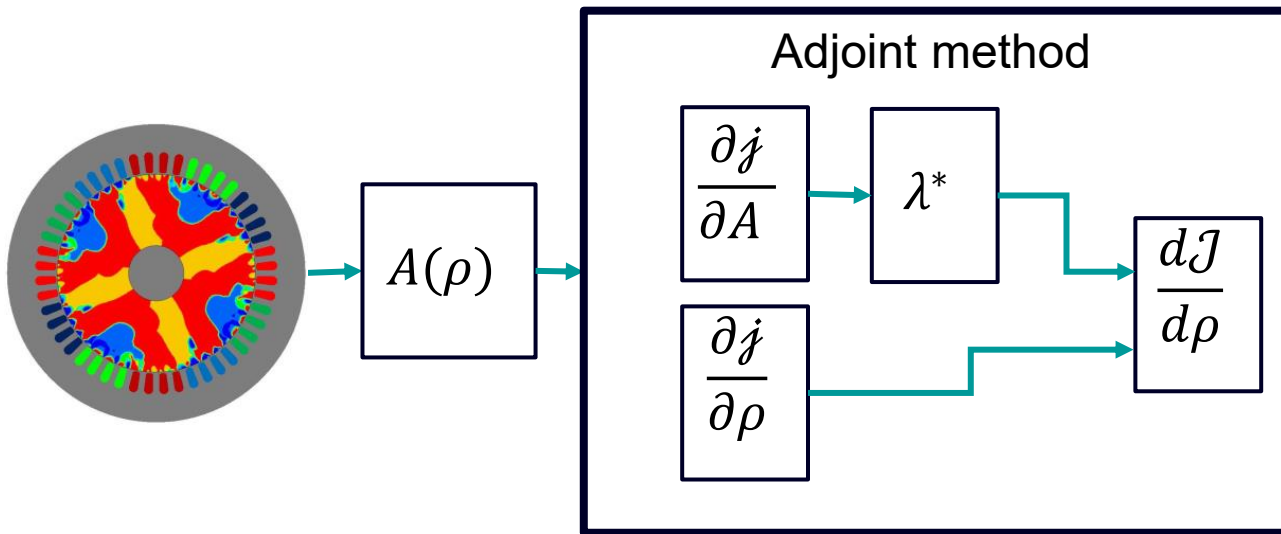
Compute $\frac{d\mathcal{J}}{d\rho}$

Implicit dependency through a PDE:

$$A = A(\rho)$$

$$\frac{d\mathcal{J}}{d\rho}(d) = \int_{\Omega} \frac{\partial j}{\partial \rho}(\rho, A) \cdot d d\Omega - \int_{\Omega} d \cdot g'(\rho)(\nu - \nu_0) \mathbf{B}^T \nabla \times \lambda d\Omega$$

$$\forall u, \int_{\Omega} (\nabla \times \lambda)^T \frac{d\mathbf{H}}{d\mathbf{B}} \nabla \times u d\Omega = \int_{\Omega} \frac{\partial j}{\partial A}(\rho, A) \cdot u d\Omega$$



- **Very efficient**
- **Requires specific terms**
- **New to the Adjoint method ? Wait for Tutorial 1 tomorrow !**

Concrete optimization criteria for electrical machines

Average Torque

$$\frac{1}{N} \sum_{i=1}^N \mathcal{T}_i$$

Torque Ripple

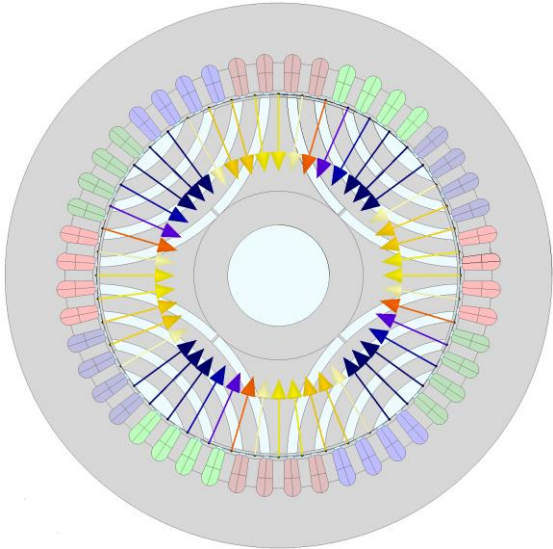
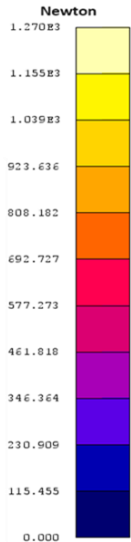
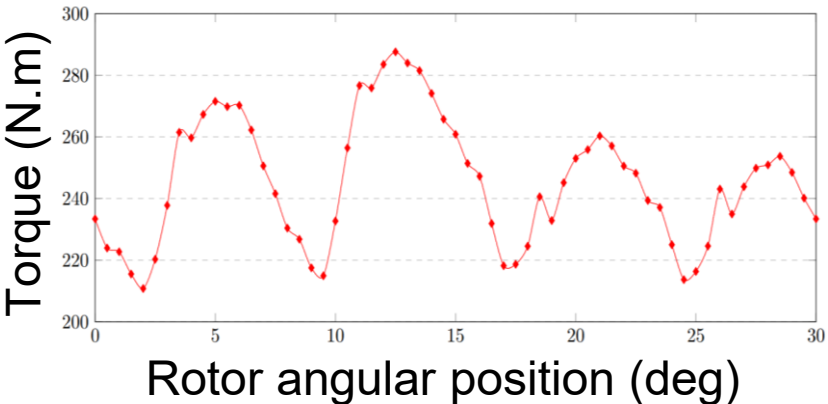
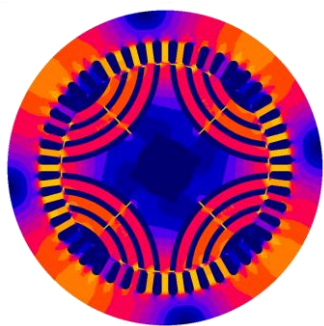
$$\frac{\mathcal{T}_{max} - \mathcal{T}_{min}}{\mathcal{T}_{avg}}$$

Torque Harmonics

$$FFT(\mathcal{T}_1, \dots, \mathcal{T}_N)$$

Force

- Global forces (actuators)
- Local force distribution



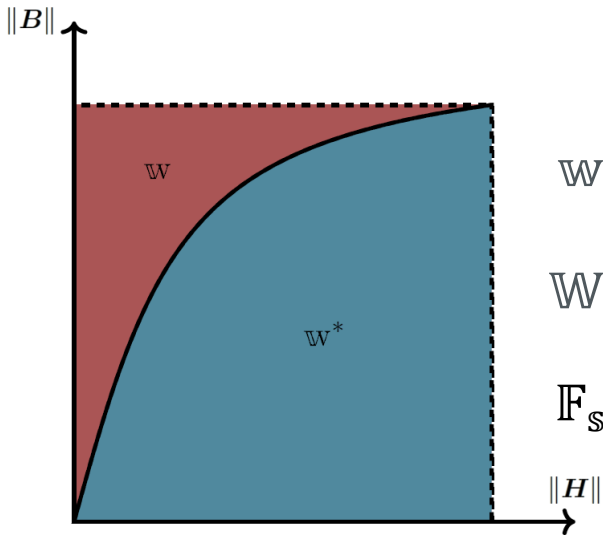
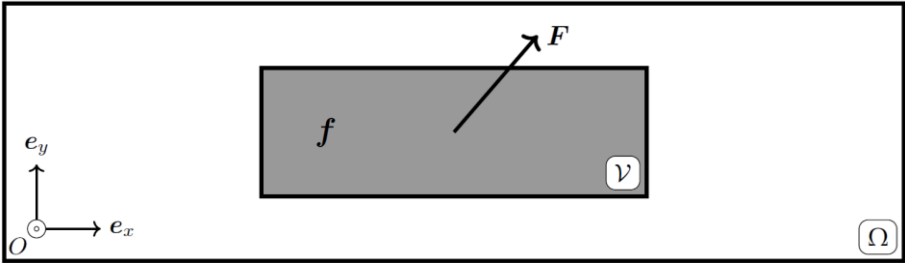
Force density in airgap

Numerical computation methods of electromagnetic forces

Maxwell Sress Tensor	The virtual work method
• Integration of Maxwell tensor components • Widely used • Easier implementation • Well known in the optimization literature	• Derivative of the magnetic energy • More complex • More accurate • Native method in Flux • New in the TO literature • Unified approach to a wide range of optimization criteria

$$\mathbb{T}_{ij} = H_i B_j - \frac{\delta_{ij}}{2} \sum_{k=1}^3 H_k B_k$$

Maxwell tensor components



$$w = \left(\int_0^B H(b) db \right)$$

$$W = \int_{\Omega} w \, d\Omega$$

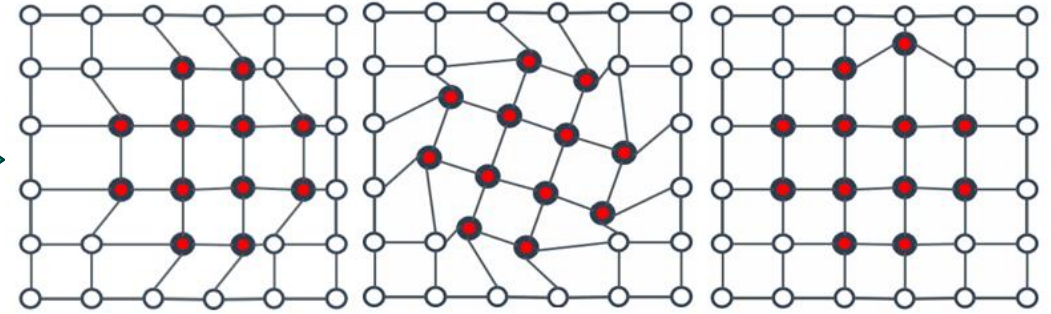
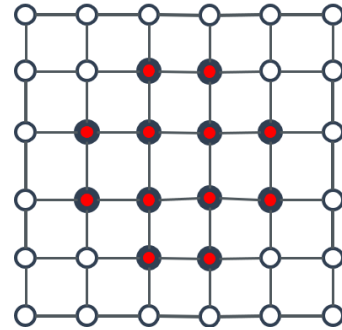
$$F_s = - \frac{\partial W}{\partial s} \Big|_{A=cste}$$

Relation between force and magnetic energy

The virtual work method – A unified approach to magnetic torque / forces / local force optimization

Discrete approach

$$\mathbb{F}_s = - \left. \frac{\partial \mathcal{W}}{\partial s} \right|_{A=cste}$$



$\mathbb{F} \cdot \mathbf{e}_x$

Torque $\mathcal{T}$

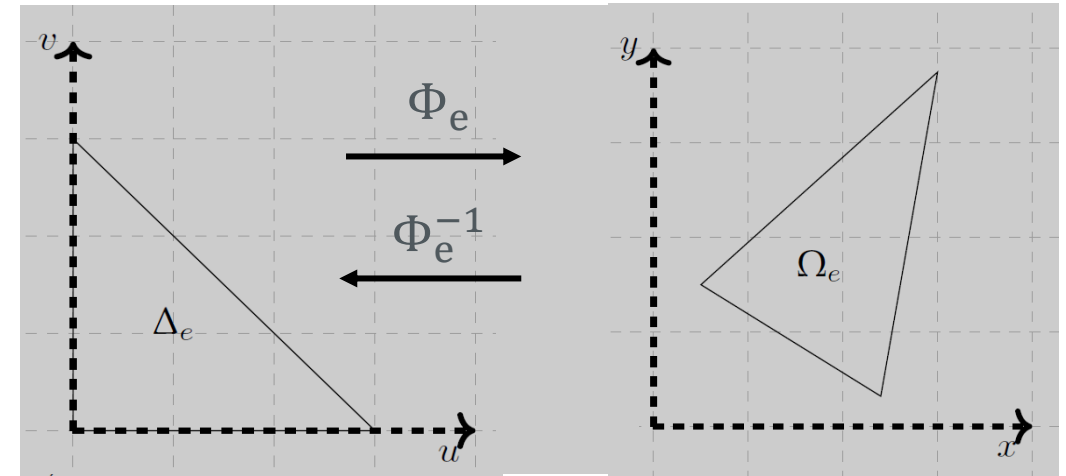
$\mathbb{f}_i \cdot \mathbf{e}_y$

Generalized force $\mathbb{F}_s: s = \{N_s, T_s\}$

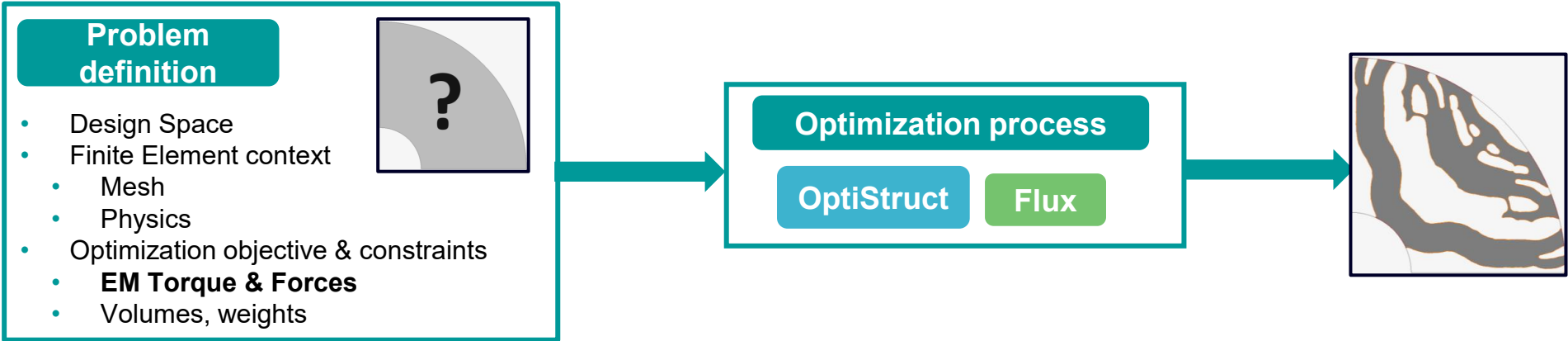
- N_s : « mobiles nodes »
- T_s : displacement

$$F_s = \sum_{e=1}^{E_\Omega} \int_{\Omega_e} \frac{1}{|\nabla \Phi_e|} \frac{\partial |\nabla \Phi_e|}{\partial s} \mathbb{W}^* - \mathbf{B}^T \nabla \Phi_e^{-1} \frac{\partial \nabla \Phi_e}{\partial s} \mathbf{H} d\Omega$$

Integrated to the adjoint method process: $\frac{\partial F_s}{\partial \rho}, \frac{\partial F_s}{\partial A}$



Synthesis –Topology optimization of Electrical Machines

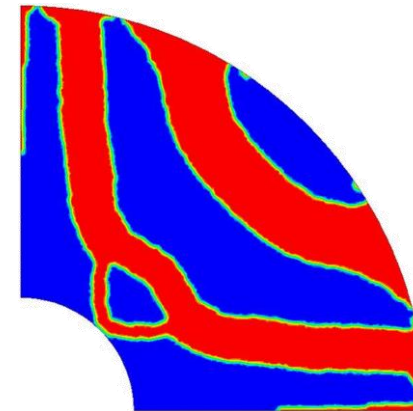


Magnetic Criterion	Integration in the optimization process	Code maturity
Torque	Yes	Flux 2D ≥ 2023
Global force	Yes	Flux 2D ≥ 2023
Force distribution	Yes	Flux development version

Adding mechanical constraints

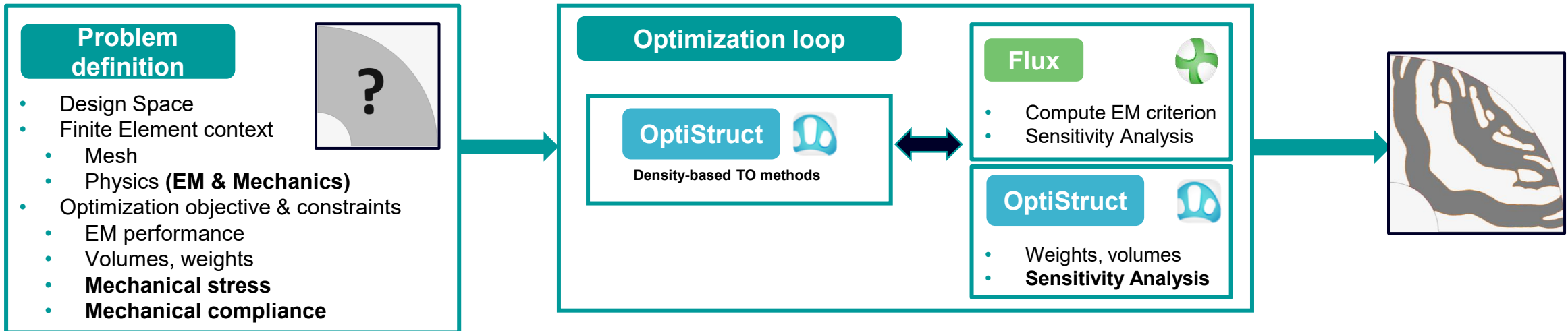
Pure magnetic optimization leads to unrealistic designs

OptiStruct is a mechanical optimization software !



MATERIAL
AIR

Magneto-mechanical topology optimization

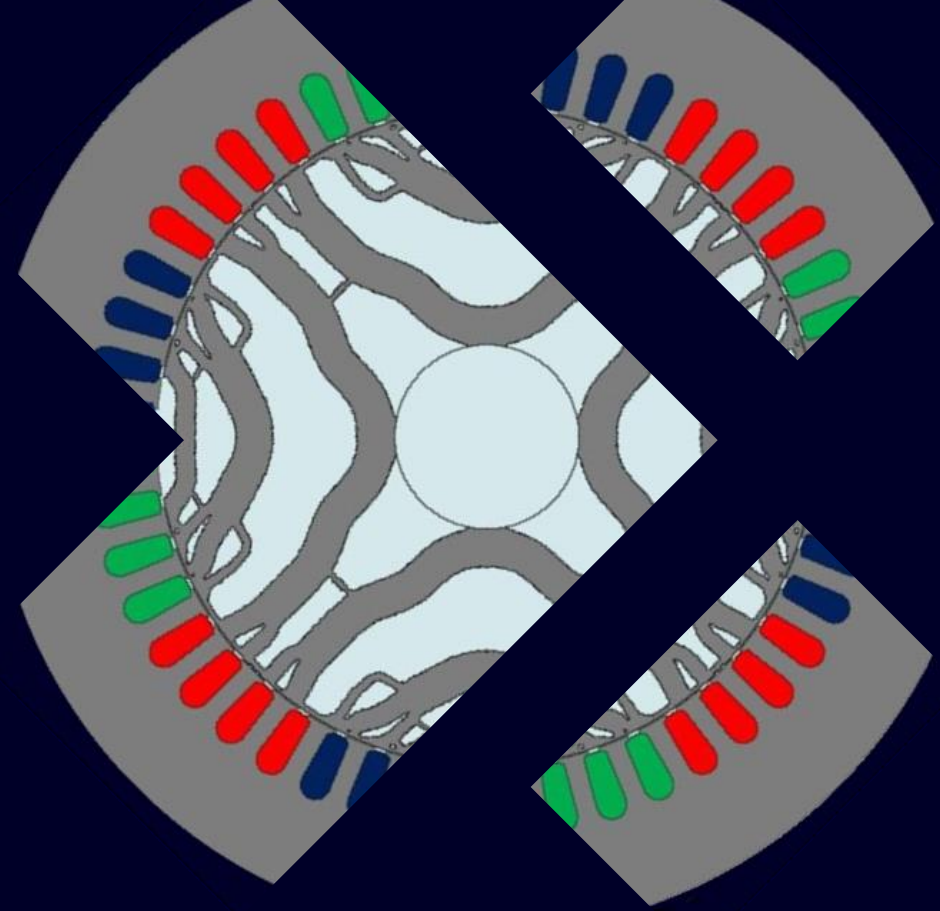


Condition 1: Single mesh of study

Condition 2: No EM/Mechanical interaction

Results

Optimization examples



Optimization of a rotor of Synchronous Reluctance Machine (1/2)

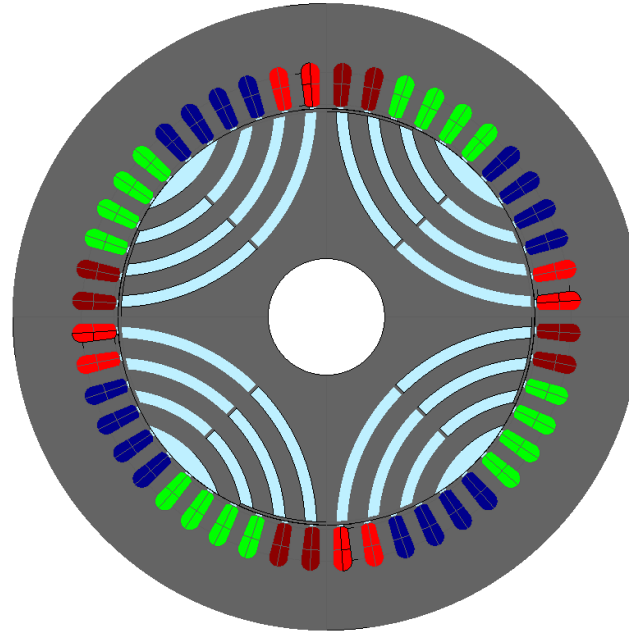
Reference Machine

- Average torque T_{avg}^0
- Torque ripple $T_r^0 = 100 \times \frac{T_{max}^0 - T_{min}^0}{T_{avg}^0}$
- Rotor material volume $V^0 = 100\%$
- Static mechanical analysis under centrifugal load

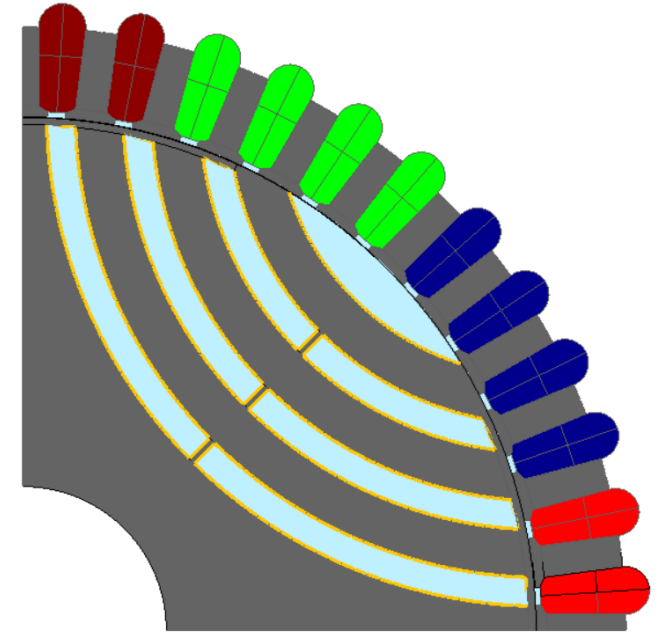
Min Rotor Volume

$$T_{avg} \geq T_{avg}^0$$

$$\sigma_{VM} \leq 80\% \sigma_{VM,max}$$



Full view on the reference machine



Zoom on the quarter of the geometry

Optimization of a rotor of Synchronous Reluctance Machine (2/2)

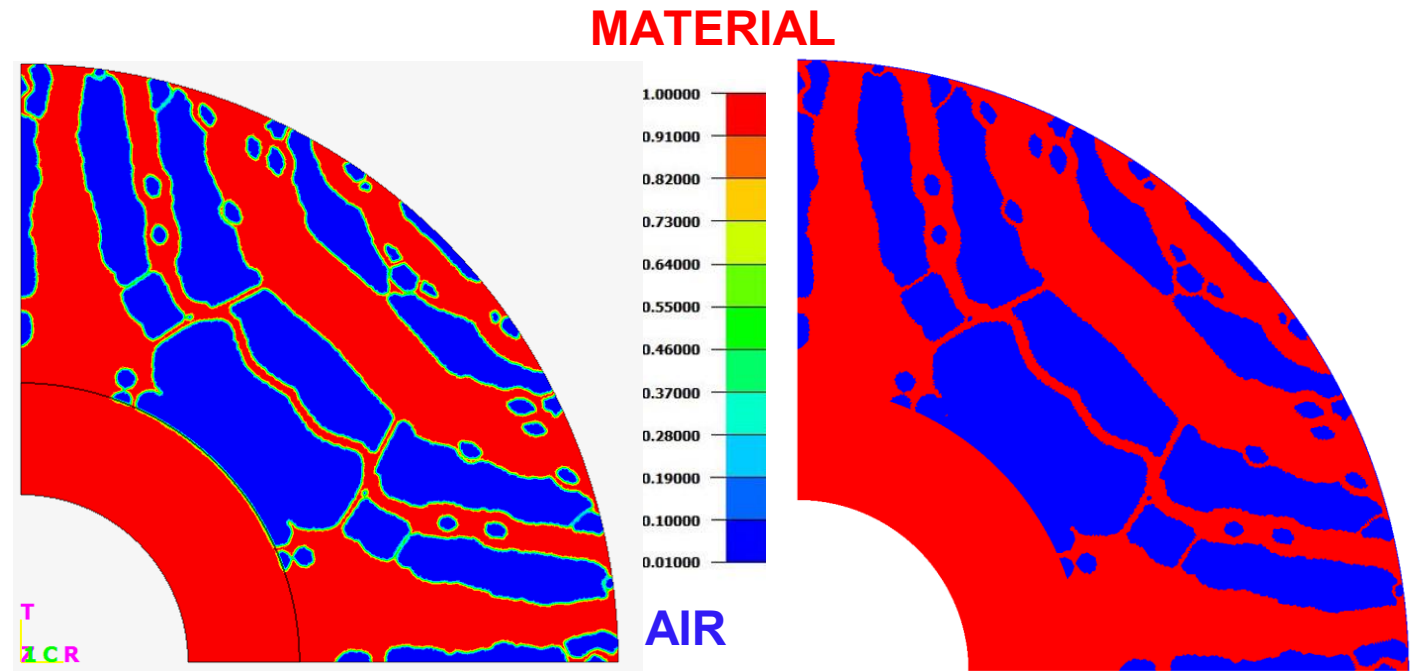
Min Volume

$$T_{avg} \geq T_{avg}^0$$

$$\sigma_{VM} \leq 80\% \sigma_{VM,max}$$

Final performance

- $T_{avg}^f = T_{avg}^0$
- $T_r^f = T_r^0$
- $V^f = 83\% V^0$
- $\sigma_{max}^f = 80\% \sigma_{max}$



Density Evolution

Final Design

Optimization of a rotor of a NISSAN Leaf motor model (1/2)

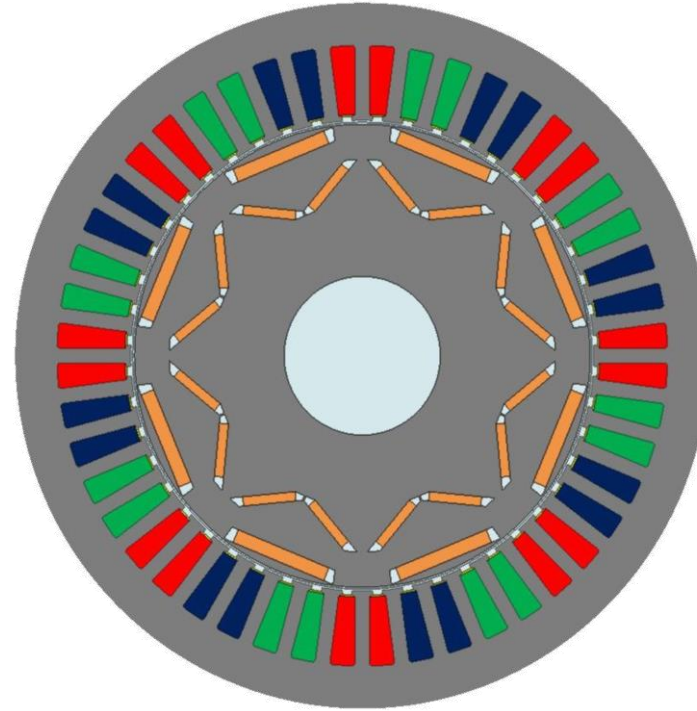
Reference Machine

- Average torque $T_{avg}^0 = 109 \text{ N.m}$
- Torque ripple $T_r^0 = 100 \times \frac{T_{max}^0 - T_{min}^0}{T_{avg}^0}$
- Rotor material volume $V^0 = 100\%$
- Static mechanical analysis under centrifugal load

Min Compliance

$$T_{avg} \geq T_{avg}^0$$

$$V \leq 70\% V_0$$



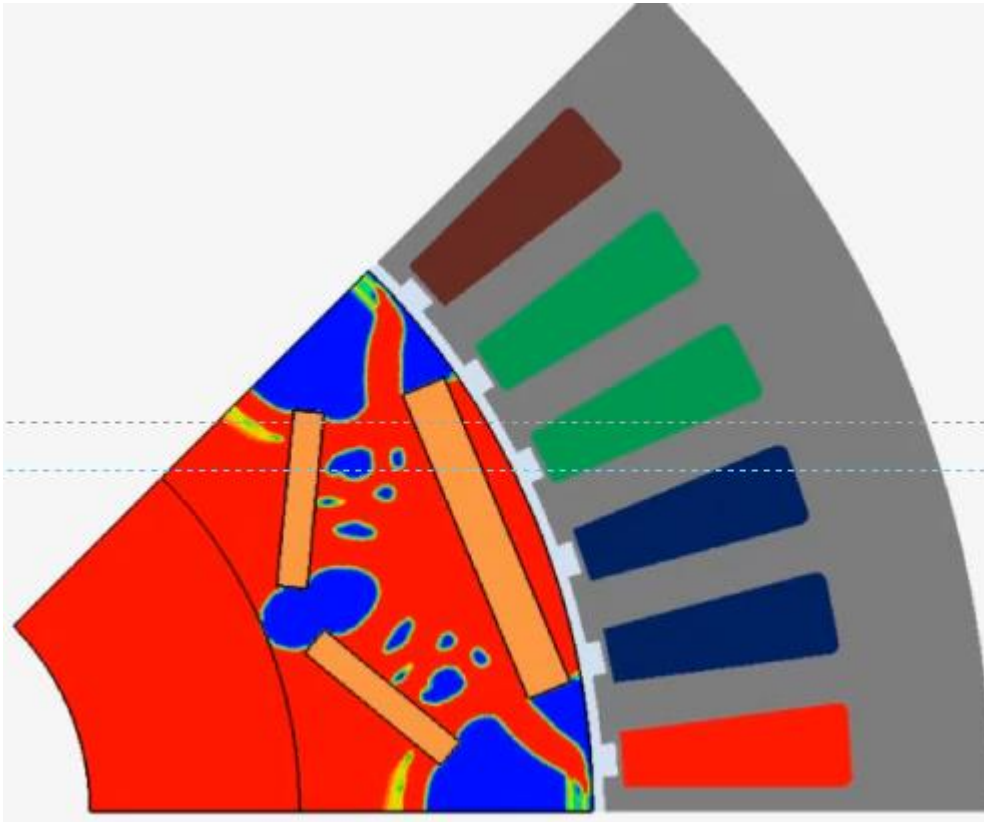
Full view on the reference machine

Optimization of a rotor of a NISSAN Leaf motor model (2/2)

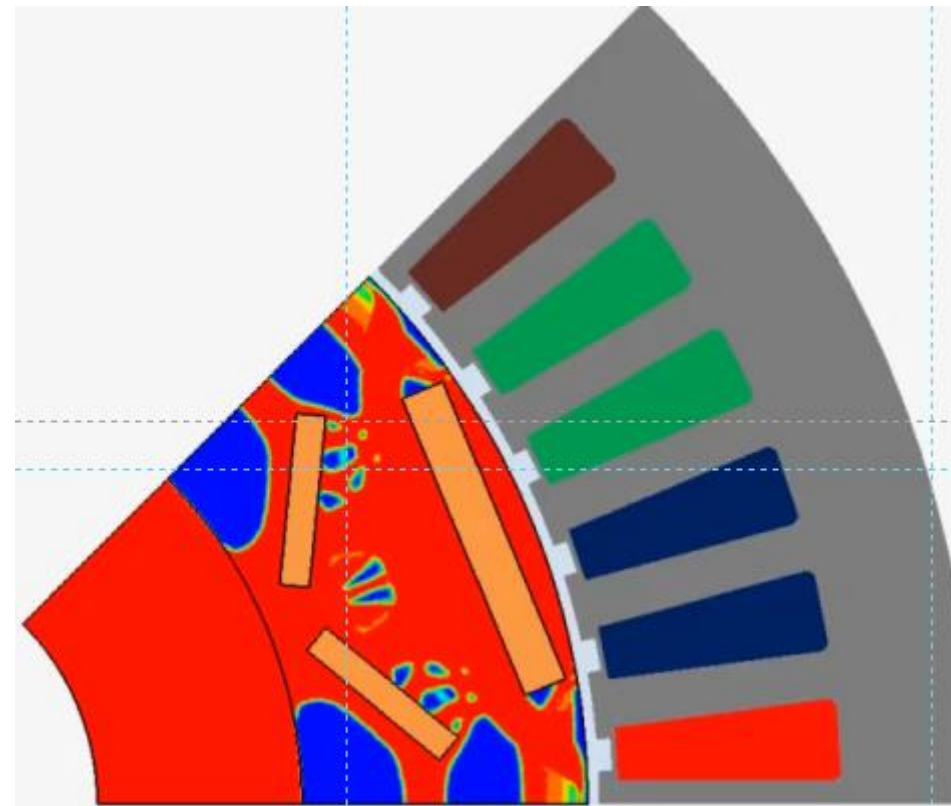
Min Compliance

$$T_{avg} \geq T_{avg}^0$$

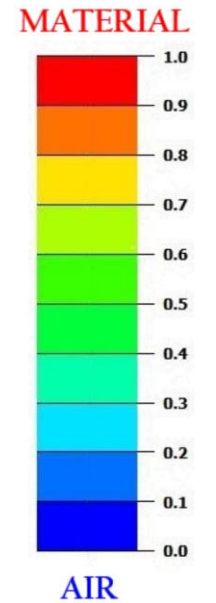
$$V \leq 70\% V_0$$



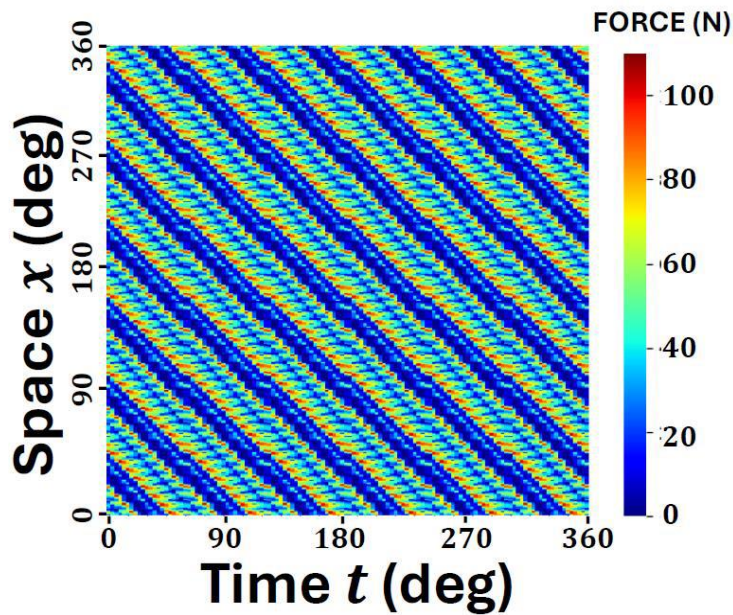
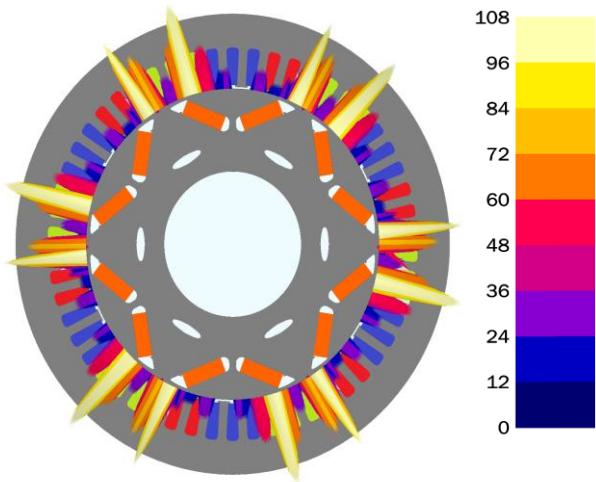
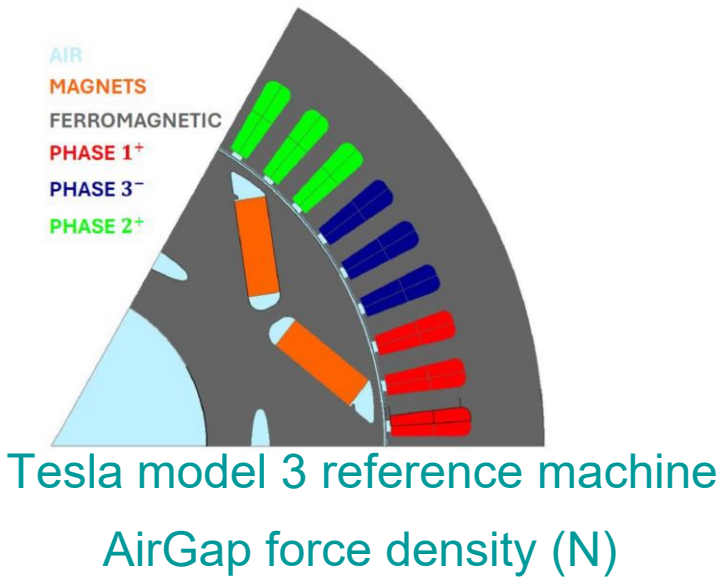
Linear mechanical analysis



Nonlinear mechanical analysis
+ contact formulation



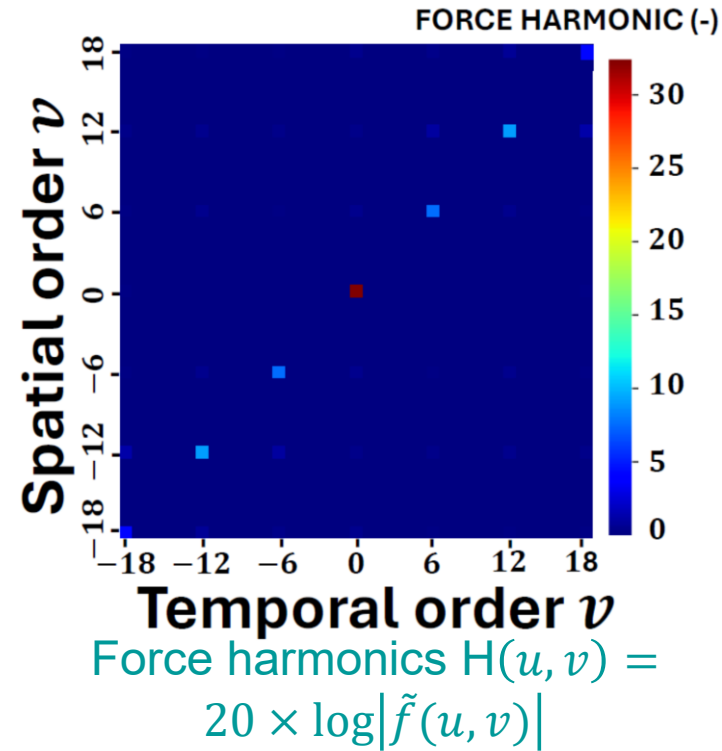
Optimization of a rotor of Tesla Model 3 motor model (1/3)



Complete force map $f(x, t)$

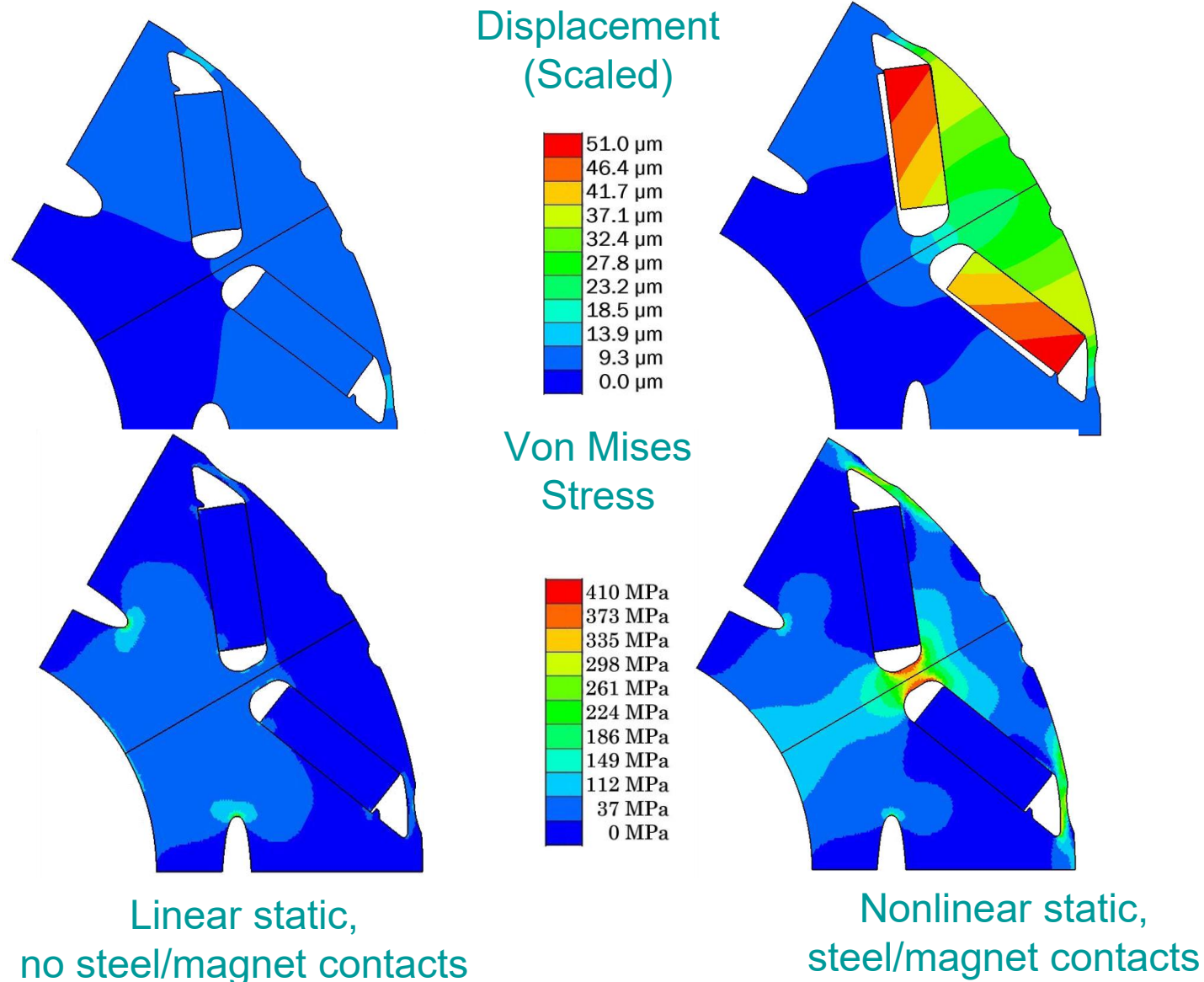
Min $H(6,6)$

$$T_{avg} \geq T_{avg}^0$$
$$\sigma_{VM} \leq \sigma_{VM,max}$$



Optimization of a rotor of Tesla Model 3 motor model (2/3)

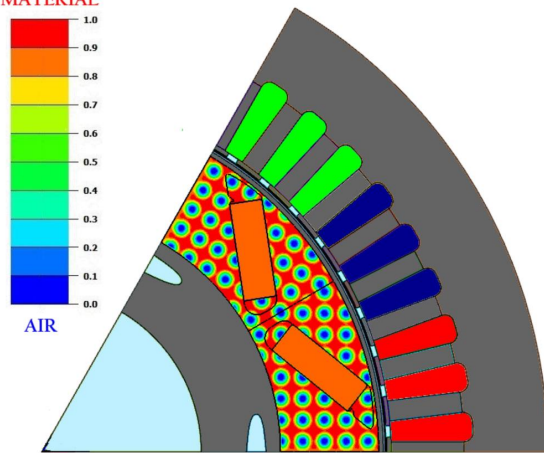
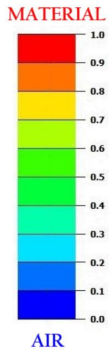
- Static centrifugal load at 18500 rpm
- Crucial effect of the mechanical analysis



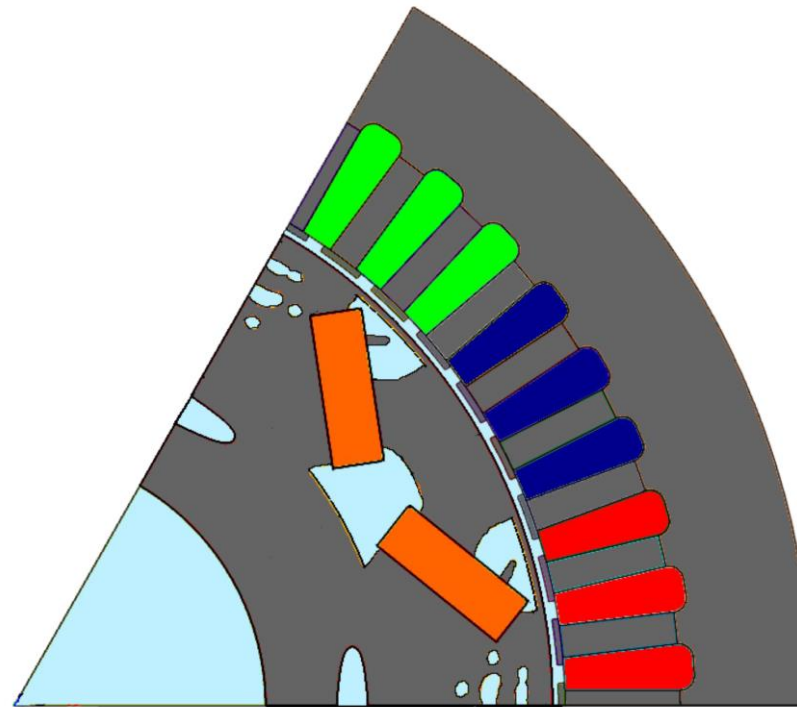
Optimization of a rotor of Tesla Model 3 motor model (3/3)

Min Harmonic

$$T_{avg} \geq T_{avg}^0$$
$$\sigma_{VM} \leq \sigma_{VM,max}$$

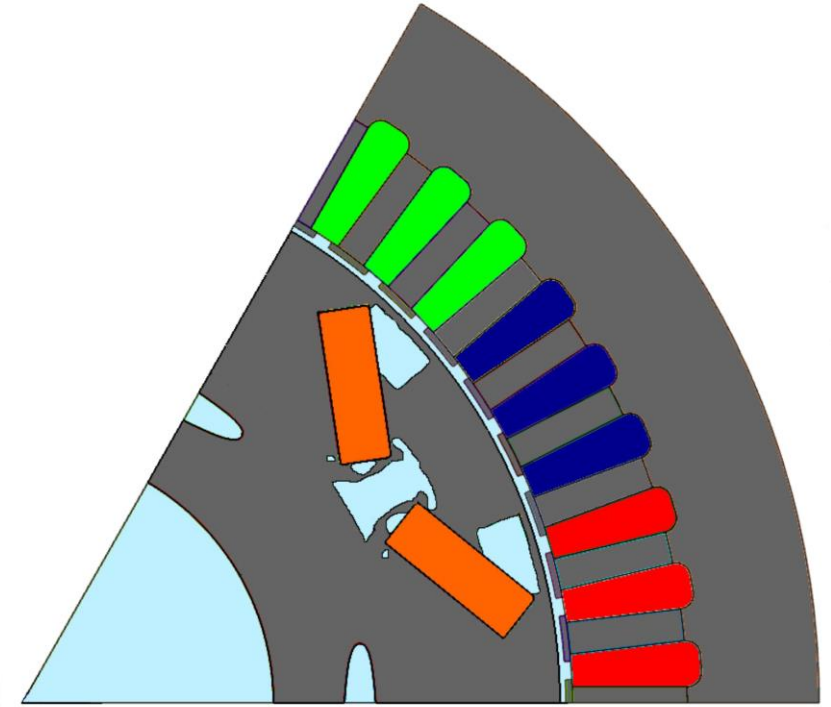


Design space & initial density



Final Design – linear mechanical analysis

- Harmonic reduced by 45%

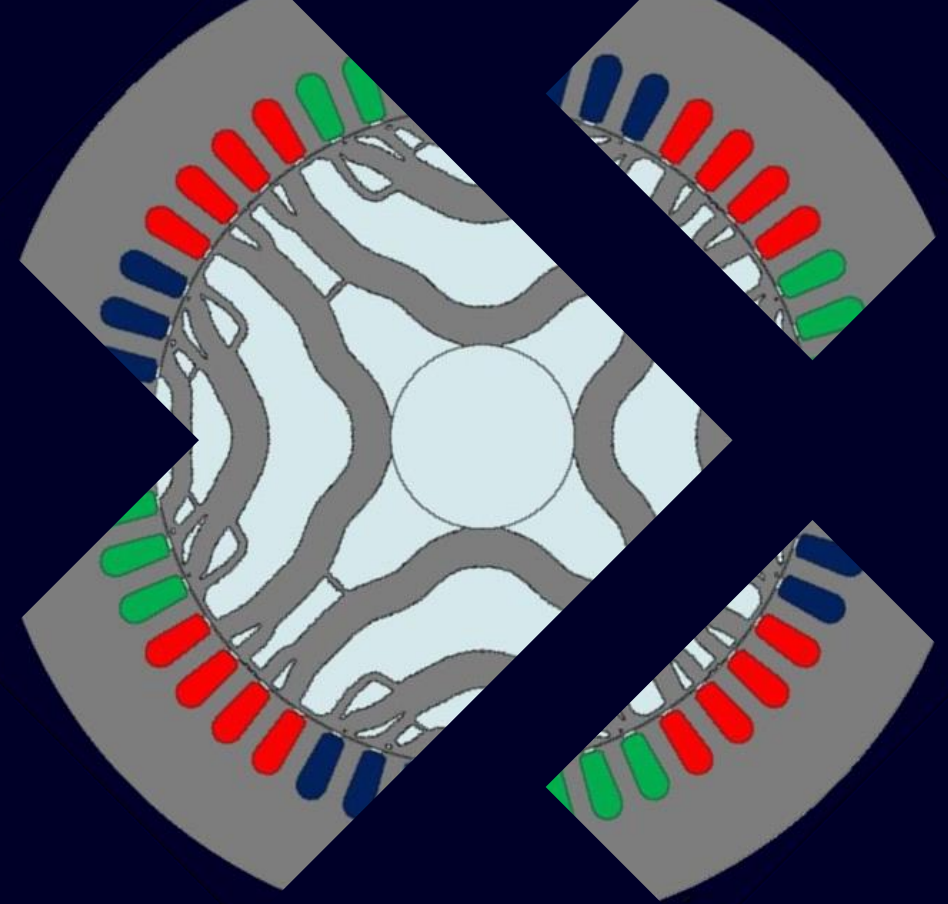


Final Design – nonlinear mechanical analysis

- Harmonic reduced by 30%
- Improved realism

SimCenter SimLab

Next-generation interface for topology optimization of electrical machines



What is SimLab ?



SimCenter
SimLab

Next generation user interface for Siemens (former Altair) solvers

Achieve complex studies from CAD to
Post-processing

Multiphysics platform for simulation engineers

New Topology Optimization **user interface** for
electrical machines. Focus on magneto-
mechanical optimization.

Thank You

